



PRODUCT REQUIREMENT DOCUMENT, PRD.

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Product Requirements Document (PRD)

Project Z — Real-Time Biometric Attendance & Workforce Management System

1. Executive Summary

Product Name

Project Z

Product Type

Enterprise-grade biometric attendance and workforce management platform.

Primary Goal

Create a centralized real-time attendance system that integrates with existing biometric devices (RONASOFT / ZKTeco-compatible terminals running `ZMM220_TFT`) to:

- capture attendance events in real time
- manage employee attendance across multiple offices/departments
- provide HR/Admin dashboards
- generate reports and analytics
- support future payroll and workforce automation

2. Problem Statement

Current attendance handling depends on:

- isolated biometric devices
- manual attendance exports

- fragmented records
- no centralized monitoring
- no realtime visibility

The organization requires:

- centralized attendance visibility
- multi-office support
- realtime attendance updates
- historical attendance analytics
- scalable workforce tracking

3. Product Vision

Project Z will become the organization's centralized workforce attendance platform.

The biometric devices will function as:

Authentication Terminals

while Project Z becomes:

The authoritative attendance and HR intelligence platform

4. Core Objectives

MVP Objectives

Realtime Attendance

Receive attendance events instantly from biometric devices via ADMS.

Centralized Employee Records

Create a unified employee database.

Multi-Device Support

Support many biometric terminals across departments/offices.

Attendance Dashboard

Provide realtime HR monitoring.

Historical Reporting

Store and analyze attendance history.

5. Technical Decisions (Chosen)

The following architecture choices are finalized for efficiency, scalability, maintainability, and compatibility with the current hardware ecosystem.

6. Final Technical Architecture

Backend Framework

Selected:

FastAPI (Python)

Reason:

- excellent async performance
- ideal for realtime APIs
- simple ADMS integration
- native WebSocket support
- excellent SDK ecosystem for ZKTeco (pyzk)
- scalable and production-proven

Frontend Framework

Selected:

React + TypeScript + TailwindCSS

Reason:

- scalable enterprise frontend
- realtime dashboard support
- excellent component ecosystem
- maintainable long-term

Database

Selected:

PostgreSQL

Reason:

- enterprise-grade
- excellent relational integrity
- ideal for attendance analytics
- supports future scaling

Realtime Layer

Selected:

WebSockets (FastAPI native)

Reason:

- realtime dashboard updates
- lightweight architecture
- simpler than polling

Device Communication

Selected:

ADMS Push + ZK SDK Hybrid Architecture

Reason:

Best long-term architecture.

ADMS:

Used for:

- realtime attendance push

SDK (Port 4370):

Used for:

- user synchronization
- device management
- future centralized enrollment

Deployment Strategy

Selected:

Dockerized Services

Reason:

- reproducible deployments
- environment consistency
- scalability

7. Hardware Ecosystem

Supported Devices

Initial Support

- RONASOFT
- ZKTeco-compatible terminals
- ZMM220_TFT platform devices

Communication Modes

Primary

ADMS Push

Device → Backend HTTP Push

Secondary

SDK Polling/Management

Backend ⇌ Device (TCP 4370)

8. Final System Architecture

Biometric Devices



ADMS Attendance Push



FastAPI Attendance Receiver



PostgreSQL Database



Realtime WebSocket Layer



React HR Dashboard

9. Core Modules

9.1 Device Management Module

Features

- register biometric devices
- track device IPs
- track serial numbers
- map devices to offices/departments
- monitor connectivity status

9.2 Employee Management Module

Features

- centralized employee records
- employee profile management
- department assignment
- office assignment
- shift assignment

9.3 Attendance Engine

Features

- realtime attendance ingestion
- attendance parsing
- duplicate prevention
- IN/OUT detection
- late arrival calculation
- overtime detection

9.4 Realtime Dashboard

Features

- currently present employees
- late employees
- absent employees
- live attendance feed
- office/department filters

9.5 Reporting Module

Features

- daily reports
- monthly summaries
- department analytics
- attendance exports
- overtime reports

10. Multi-Office Architecture

Design Decision

Devices represent locations.

NOT employee ownership.

Example

Device	Office	Department
172.16.40.12	HQ Floor 1	IT
172.16.40.13	HQ Floor 2	HR

11. Employee Identity Strategy

Selected Strategy

Reuse Existing Device Users

Reason

- avoids re-enrollment
- preserves fingerprint data
- reduces migration complexity

Backend Employee Identity

Backend becomes authoritative.

Device IDs are mapped into backend employees.

12. Fingerprint Strategy

Selected Strategy

Device-Managed Biometrics

Meaning

Fingerprint templates remain on devices initially.

Backend manages:

- employee metadata
- attendance logic

- reporting

Reason

Most efficient MVP approach.

Avoids:

- template synchronization complexity
- vendor-specific biometric formats

13. Attendance Flow

Employee scans fingerprint

↓

Device validates fingerprint

↓

Device pushes attendance via ADMS

↓

FastAPI receives event

↓

Attendance saved to PostgreSQL

↓

WebSocket broadcasts update

↓

HR dashboard updates instantly

14. Network Design

Current LAN Design

Devices

172.16.40.12

172.16.40.13

...

Backend Server

Example:

172.16.40.50:8081

15. API Design

Core Endpoints

Attendance Receiver

POST /iclock/cdata

Employees

GET /employees

POST /employees

PUT /employees/{id}

Attendance

GET /attendance/live

GET /attendance/history

Devices

GET /devices

POST /devices

16. Database Design

Core Tables

employees

Field	Type
id	UUID
employee_code	VARCHAR
name	VARCHAR
department_id	FK
shift_id	FK

devices

Field	Type
id	UUID
serial_number	VARCHAR
ip_address	VARCHAR
office_id	FK
department_id	FK

attendance_logs

Field	Type
id	UUID
employee_id	FK
device_id	FK
timestamp	TIMESTAMP
verify_type	INTEGER

departments

Field	Type
id	UUID
name	VARCHAR

shifts

Field	Type
id	UUID
start_time	TIME
end_time	TIME

17. Authentication & Roles

Roles

Role	Permissions
Super Admin	Full system control
HR Admin	Employee + attendance management
Department Manager	Department-only visibility
Viewer	Read-only reports

18. Security Requirements

Required Security Measures

Network

- devices remain on private LAN/VPN
- never expose port 4370 publicly

API

- JWT authentication
- RBAC permissions
- HTTPS in production

Database

- encrypted backups
- audit logging

19. Scalability Strategy

Project Z must support:

- multiple offices
- multiple devices
- thousands of employees
- future cloud deployment

20. Future Features (Post-MVP)

Phase 2

- centralized user provisioning
- device synchronization
- leave management
- payroll integration

Phase 3

- mobile application
- geo-fencing
- facial recognition

- SaaS multi-tenant architecture

21. MVP Scope

Included

- ✓ realtime attendance
- ✓ ADMS integration
- ✓ employee management
- ✓ attendance dashboard
- ✓ reporting
- ✓ multi-device support
- ✓ multi-office support

Excluded From MVP

- ✗ centralized fingerprint storage
- ✗ advanced payroll engine
- ✗ mobile apps
- ✗ facial recognition
- ✗ cloud multi-tenancy

22. Final Architectural Decisions

Category	Decision
Backend	FastAPI
Frontend	React + TypeScript
Database	PostgreSQL

Realtime	WebSockets
Device Integration	ADMS + SDK Hybrid
Fingerprint Storage	Device-managed
Deployment	Docker
Authentication	JWT
Network	LAN/VPN
Attendance Model	Centralized backend

23. Recommended Development Phases

Phase 1 — Infrastructure

- FastAPI setup
- PostgreSQL setup
- Docker environment

Phase 2 — Device Integration

- ADMS receiver
- attendance parsing
- device registration

Phase 3 — Core Backend

- employee management
- attendance engine
- reporting APIs
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Phase 4 — Frontend

- HR dashboard
- realtime attendance UI
- reporting interface

Phase 5 — Production Hardening

- authentication
- logging
- monitoring
- backups

24. Success Criteria

Project Z succeeds when:

- attendance appears in realtime
- HR can monitor all offices centrally
- attendance history is reliable
- devices synchronize reliably
- reports are accurate
- system scales without architectural rewrites

25. Final Recommendation

The chosen architecture is intentionally:

- scalable
- maintainable
- enterprise-ready
- compatible with existing biometric hardware
- optimized for realtime attendance workflows

The system design minimizes:

- biometric complexity
- vendor lock-in
- migration risk
- operational overhead

while maximizing:

- realtime visibility
- future scalability
- deployment speed
- maintainability

Final Engineering Conclusion

Project Z should treat biometric devices as:

distributed authentication terminals

while the backend becomes:

the centralized workforce intelligence platform

This architecture is the most efficient, scalable, and production-safe design for your environment.

